

DATABASE SYSTEMS ENGINEERING

DDL AND DML OPERATIONS

Project: Doctor Appointment Booking System

Database: appointment_db

Aim

To perform DDL and DML operations using MySQL for a Doctor Appointment Booking System.

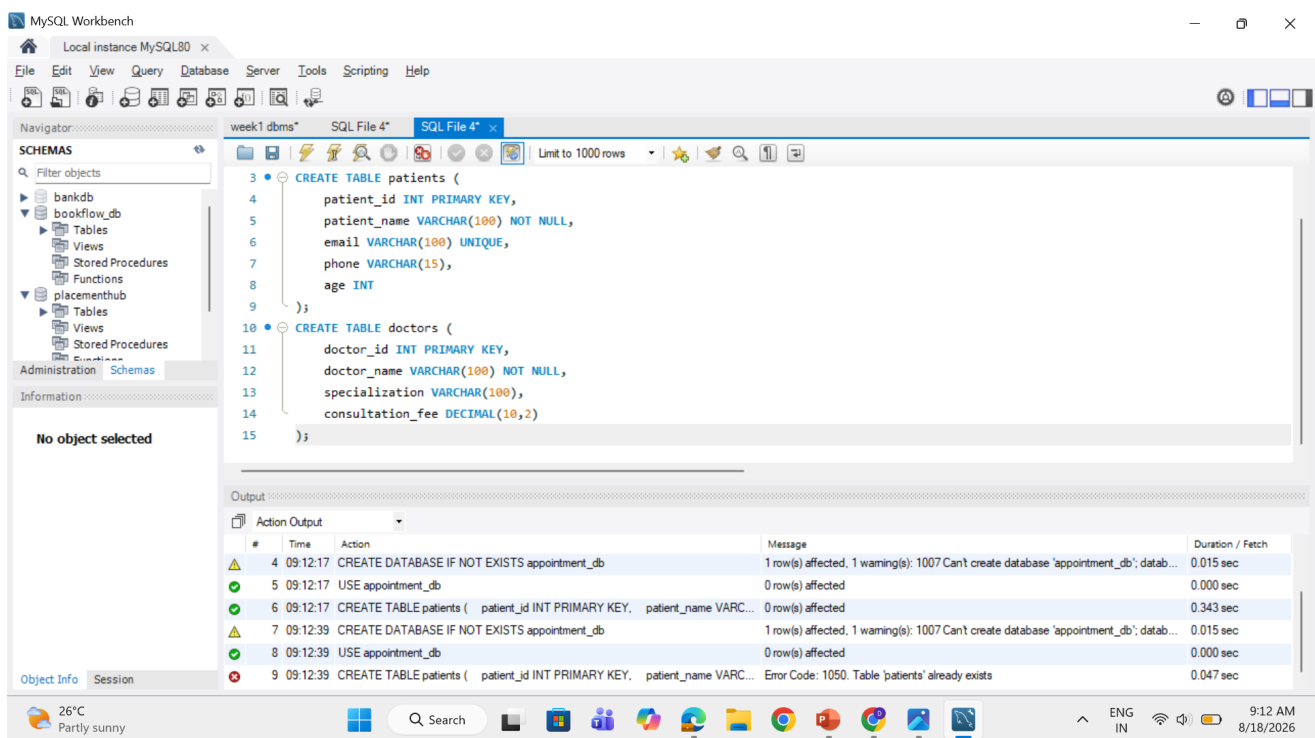
Database Structure

The practical uses the **appointment_db** database with **patients** and **doctors** tables.

1. CREATE DATABASE

SQL:

```
CREATE DATABASE IF NOT EXISTS appointment_db;  
USE appointment_db;
```



MySQL Workbench execution/result screenshot.

2. CREATE TABLE

SQL:

```
CREATE TABLE patients (  
    patient_id INT PRIMARY KEY,  
    patient_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE,  
    phone VARCHAR(15),  
    age INT  
);
```

```

CREATE TABLE doctors (
doctor_id INT PRIMARY KEY,
doctor_name VARCHAR(100) NOT NULL,
specialization VARCHAR(100),
consultation_fee DECIMAL(10,2)
);

```

The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'Schemas' panel with 'appointment_db' selected. The main editor window contains two SQL queries: a 'CREATE TABLE patients' statement and a 'CREATE TABLE doctors' statement. The 'Output' panel at the bottom shows the execution results of these queries.

#	Time	Action	Message	Duration / Fetch
8	09:12:39	USE appointment_db	0 row(s) affected	0.000 sec
9	09:12:39	CREATE TABLE patients (patient_id INT PRIMARY KEY, patient_name VARCHAR...	Error Code: 1050. Table 'patients' already exists	0.047 sec
10	09:14:30	CREATE DATABASE IF NOT EXISTS appointment_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'appointment_db'; datab...	0.016 sec
11	09:14:30	USE appointment_db	0 row(s) affected	0.015 sec
12	09:14:30	CREATE TABLE patients (patient_id INT PRIMARY KEY, patient_name VARCHAR...	Error Code: 1050. Table 'patients' already exists	0.016 sec
13	09:16:36	CREATE TABLE doctors (doctor_id INT PRIMARY KEY, doctor_name VARCHAR...	0 row(s) affected	0.125 sec

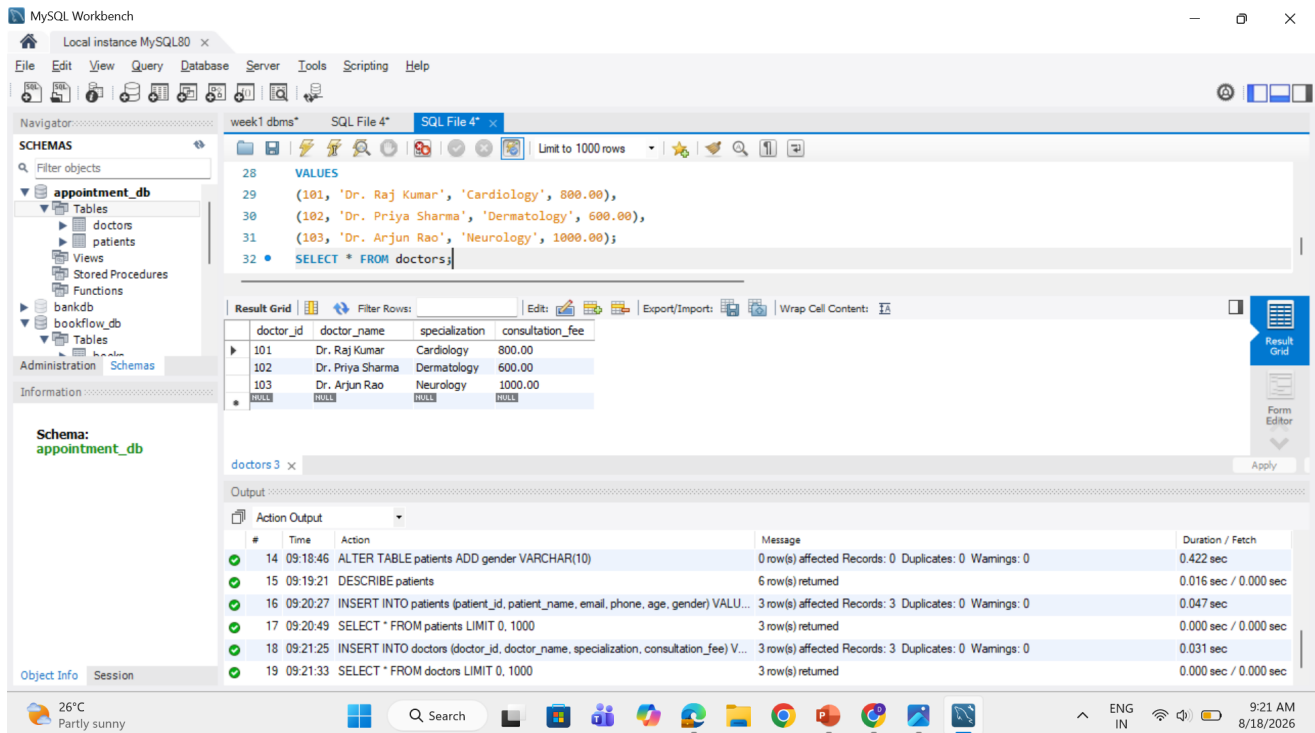
MySQL Workbench execution/result screenshot.

3. ALTER TABLE

SQL:

```
ALTER TABLE patients  
ADD gender VARCHAR(10);
```

```
DESCRIBE patients;
```



MySQL Workbench execution/result screenshot.

4. INSERT

SQL:

```
INSERT INTO patients  
(patient_id, patient_name, email, phone, age, gender)  
VALUES  
(1, 'Deeraj', 'deeraj@gmail.com', '9876543210', 19, 'Male'),  
(2, 'Rahul', 'rahul@gmail.com', '9876543211', 20, 'Male'),  
(3, 'Ananya', 'ananya@gmail.com', '9876543212', 21, 'Female');
```

```
INSERT INTO doctors  
(doctor_id, doctor_name, specialization, consultation_fee)  
VALUES  
(101, 'Dr. Raj Kumar', 'Cardiology', 800.00),  
(102, 'Dr. Priya Sharma', 'Dermatology', 600.00),  
(103, 'Dr. Arjun Rao', 'Neurology', 1000.00);
```

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

appointment_db

Tables

doctors

patients

Views

Stored Procedures

Functions

bankdb

bookflow_db

Administration Schemas

Information

Schema: appointment_db

SQL File 4*

```
33 • UPDATE doctors
34   SET consultation_fee = 700.00
35   WHERE doctor_id = 102;
36
37 • SELECT * FROM doctors;
```

Result Grid

doctor_id	doctor_name	specialization	consultation_fee
101	Dr. Raj Kumar	Cardiology	800.00
102	Dr. Priya Sharma	Dermatology	600.00
103	Dr. Arjun Rao	Neurology	1000.00
HIDE	HIDE	HIDE	HIDE

doctors 4 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 18	09:21:25	INSERT INTO doctors (doctor_id, doctor_name, specialization, consultation_fee) V...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
✓ 19	09:21:33	SELECT * FROM doctors LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
⚠ 20	09:21:58	CREATE DATABASE IF NOT EXISTS appointment_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'appointment_db'; datab...	0.000 sec
✓ 21	09:21:58	USE appointment_db	0 row(s) affected	0.000 sec
✗ 22	09:21:58	CREATE TABLE patients (patient_id INT PRIMARY KEY, patient_name VAR...	Error Code: 1050. Table 'patients' already exists	0.000 sec
✓ 23	09:21:59	SELECT * FROM doctors LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

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9:22 AM 8/18/2026

MySQL Workbench execution/result screenshot.

5. SELECT

SQL:

```
SELECT * FROM patients;  
SELECT * FROM doctors;
```

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'appointment_db' selected. The main editor shows a SQL script with the following queries:

```
37 • SELECT * FROM doctors;  
38 • DELETE FROM patients  
39 • WHERE patient_id = 3;  
40  
41 • SELECT * FROM patients;
```

The 'Result Grid' shows the results of the last query, displaying a table with 6 columns: patient_id, patient_name, email, phone, age, and gender. The data is as follows:

patient_id	patient_name	email	phone	age	gender
1	Deeraj	deeraj@gmail.com	9876543210	19	Male
2	Rahul	rahul@gmail.com	9876543211	20	Male
3	Ananya	ananya@gmail.com	9876543212	21	Female

The 'Output' pane at the bottom shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
20	09:21:58	CREATE DATABASE IF NOT EXISTS appointment_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'appointment_db'; datab...	0.000 sec
21	09:21:58	USE appointment_db	0 row(s) affected	0.000 sec
22	09:21:58	CREATE TABLE patients (patient_id INT PRIMARY KEY, patient_name VAR...	Error Code: 1050. Table 'patients' already exists	0.000 sec
23	09:21:59	SELECT * FROM doctors LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
24	09:22:19	SELECT * FROM patients LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
25	09:22:37	SELECT * FROM patients LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec

MySQL Workbench execution/result screenshot.

6. UPDATE

SQL:

```
UPDATE doctors  
SET consultation_fee = 700.00  
WHERE doctor_id = 102;  
  
SELECT * FROM doctors;
```

7. DELETE

SQL:

```
DELETE FROM patients
WHERE patient_id = 3;

SELECT * FROM patients;
```

8. DROP TABLE

SQL:

```
CREATE TABLE test_table (
  id INT
);

DROP TABLE test_table;
```

Conclusion

DDL and DML operations were successfully performed in MySQL for the Doctor Appointment Booking System. The database and tables were created, table structure was modified, records were inserted and retrieved, a record was updated, and a record was deleted. A temporary table was also used to demonstrate the DROP operation.